# Shawn Agarwal

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## Education

**University of Oklahoma** Aug 2024 – May 2027
*B.S. Electrical Engineering,* ***GPA 4.0*** *— Oklahoma Regents Scholar, Presidential Honor Roll* *Norman, OK*

## Experience

**Amazon** May 2026 – Aug 2026
*Operations Intern, OKC1 Fulfillment Center* *Oklahoma City, OK*

- Analyzed station throughput data to find bottlenecks; redesigned the replenishment procedure and support-runner role, raising pack rate **6%** (**$790K/yr** projected at OKC1).
- Supervised shifts of **280** associates alongside technicians, engineers, and area managers against daily safety, quality, and output targets.

**Center for Quantum Research & Technology, University of Oklahoma** Jan 2026 – Present
*Undergraduate Researcher, Engineering Pathways Fellow* *Norman, OK*

- Fabricated and characterized **10+** thin-film semiconductor samples, independently operating the group's vacuum deposition and X-ray diffraction equipment.
- Cross-analyzed structural, electrical, and optical data to identify a film-thickness limit degrading device performance; result redefined the group's growth parameters.
- Built **XRDC**, a MATLAB analysis tool now used group-wide, replacing hours of manual X-ray data fitting per sample.

**Heading Software** Jan 2026 – Present
*Founder & ML Engineer — REP, computer-vision lifting app (closed beta, OU Powerlifting)* *Norman, OK*

- Built the computer-vision pipeline that analyzes uploaded lift videos and returns rep count, bar path, bar speed, and form checks in **under 60 seconds**.
- Built an automated test suite over **85** labeled clips that gates every model change (**85%** exact rep-count accuracy); benchmarked on-device inference (0.7–1.7 s/frame on iPhone) and moved analysis to server GPUs to meet the 60 s target.
- Deployed the production backend (auth, rate limiting, sandboxed video processing, monitoring) and the **React Native/TypeScript** app on **Supabase**.

**Data Institute for Societal Challenges, University of Oklahoma** Sep 2024 – May 2025
*Undergraduate Researcher, Machine Learning* *Norman, OK*

- Cut human labeling **10×** on a **1.6M**-record text classification task by fine-tuning **BERT** with active learning — 95% accuracy from 100 labels vs. 90% from 1,000 with standard training.

## Hardware Projects

**Programmable Protocol-Emulator ASIC** Sep 2026 – Present
*Jane Street ASIC Competition entry — Verilog, cocotb, SymbiYosys, Tiny Tapeout*

- Designed a firmware-programmable pin engine in **Verilog** that runs UART, SPI, and $I^2C$ as programs on a 15-opcode instruction set, with host FIFOs and a timestamped capture unit for clock-exact protocol fault injection.
- Verified with **22** cocotb regressions, mutation controls, and **formal proofs**; completed layout on IHP 130 nm (10 MHz) with clean DRC/LVS and gate-level timing passing at three corners, all reproduced in CI.

**Intelligent Ground Vehicle Competition** Aug 2025 – May 2026
*Electrical Member, Sooner Competitive Robotics — Altium, STM32, RTOS, CAN*

- Designed and brought up custom **PCBs** for power regulation, signal conditioning, and safety interlocks — schematic through layout, assembly, and verification against competition safety rules.
- Wrote real-time **C/C++** firmware (STM32, RTOS) for motor control; integrated LiDAR, cameras, GPS, and emergency-stop on a shared **CAN bus**; built **OpenCAN**, a USB diagnostic tool for live bus monitoring.

**Student Tele-Operated Robotics Mission** Aug 2024 – May 2025
*Electrical Team Lead, Sooner Competitive Robotics — power electronics, embedded firmware*

- Designed the robot's power distribution system and wrote the firmware for wireless remote control.
- Led a **5-person** electrical team through design reviews, integration, and test against a fixed end-of-year mission deadline.

## Technical Skills

**Languages**: C/C++, Python, MATLAB, Verilog, TypeScript, SQL
**Hardware**: Altium (PCB design), STM32 / ARM Cortex-M, RTOS, CAN bus, cocotb, Yosys, oscilloscope & lab debug
**Software**: PyTorch, React Native, Supabase/PostgreSQL, Docker, Git
**Activities**: IEEE, Engineers' Club, Sooner Competitive Robotics, Society of Asian Scientists and Engineers